Simulation of Fluid Flow in a Tesla Valve.

Hatef Khadivinassab, Daniele Rizzi

July 2, 2020

1 Introduction

Microfluidics are widely employed in areas from biomedical and drug delivery to space and fuel cell systems. Miniaturized valves (microvalves) are microfluidic two-port components that regulates the flow between two fluidic ports. These valves are the key components in devices used in drug delivery, cardiology and handling biomolecules [1] [2]. In such applications, the ability to control and move the fluid together with the ability to predict the behavior of the flow inside the valves is of utmost importance.

Microvalves come in two major categories; active and passive. Active valves use moving parts such as mechanical moving membranes that are coupled by magnetic, electric, piezoelectric or thermal actuation methods [1]. Passive valves, on the other hand, do not incorporate any moving parts, where their operational state (open or closed) is directly determined by the fluid.

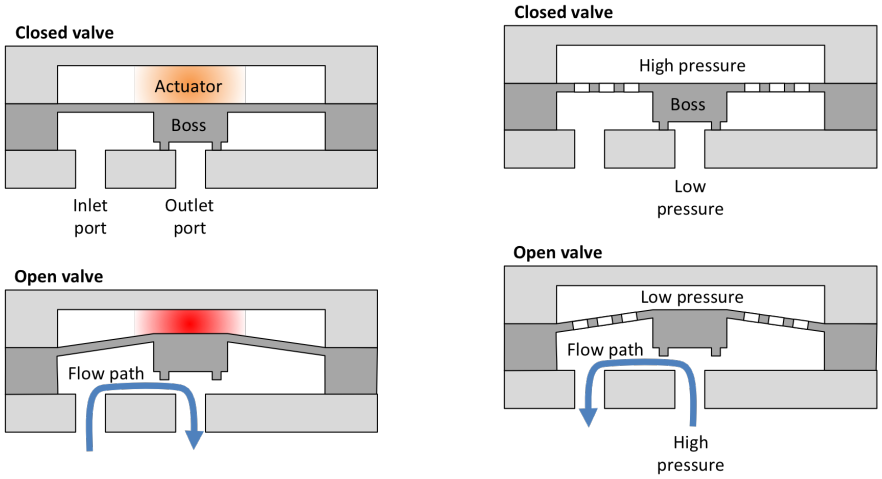


Figure 1: *Functional scheme of active (a) and passive (b) valves* [1].

Microvalves are low cost, easy to fabricate and scalable. These characteristics contribute to their increasing usage in the aforementioned industries. Despite all these advantages, their

[1]Source: `https://en.wikipedia.org/wiki/Microvalve`

operation is characterized by the equations of moving fluids in the form of the Navier-Stokes equations, whose solution is still an open problem in the field of mathematics. To predict the behavior of a moving fluid, a Computational Fluid Dynamics (CFD) simulation must be performed.

1.1 Tesla valve

Nikola Tesla patented one of the first passive valve designs. The so called Tesla valve allows flow preferentially in one direction, while prevents the flow in the other direction. The basic design of this valve consists of channel that is split in a Y junction that re-enters in a T channel, shown in Figure 2. This valve can be advantageous since it is easy to design, easy to fabricate and it can be scaled with ease.

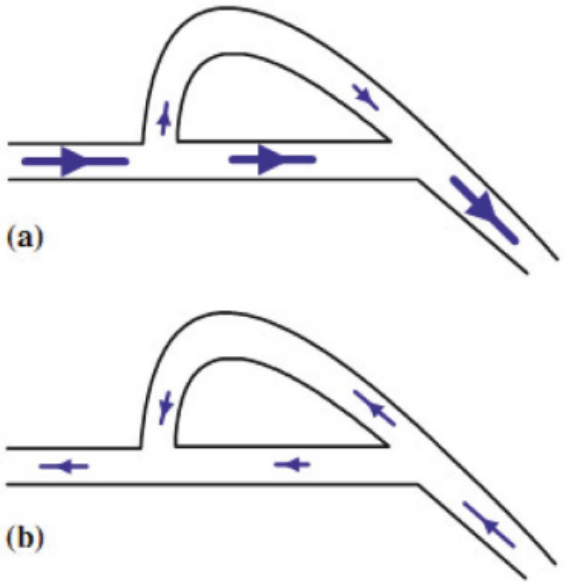


Figure 2: *Flow in a Tesla valve in forward (a) and reverse (b) direction [3].*

Such a valve, is structured to have a higher pressure drop for the flow in one direction (reverse) than the other (forward). This difference in flow resistance causes a net directional flow rate in the forward direction in oscillating flows. The efficiency is often expressed in terms of the diodicity D_i, defined as the ratio of pressure drops for identical flow rates

$$D_i = \left(\frac{\Delta p_r}{\Delta p_f} \right)_Q \tag{1}$$

where Δp_r is the reverse flow pressure drop and Δp_f the forward flow pressure drop for flow rate. Since the diodicity of such valve depends on the pressure drop between direct flow versus the reverse flow, higher diodicity values may be obtained by changing geometrical parameters of this type of valve, such as:

- Angle between channel and T junction

- Angle between channel and Y junction
- Number of branches of the valve

In this paper we will analyse and discuss the diodicity of two Tesla microvalve designs with a 3D CFD simulation using the open source computing platform FEniCS.

2 Methodology

2.1 Geometry

2.1.1 Design

The Tesla valves were designed following the instructions provided in [9]. The design variables are shown in Figure 3.

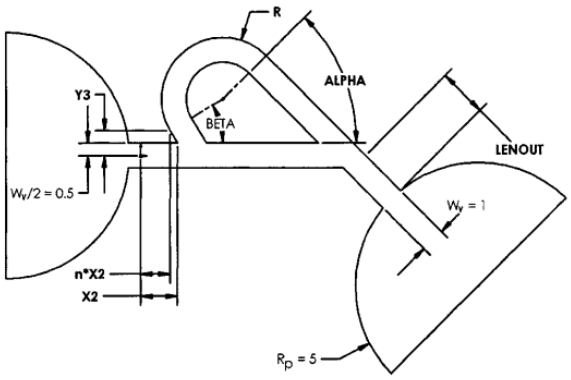


Figure 3: *Design of Tesla valve provided in [9].*

Respective dimensions for designs A (Figure 4) and B (Figure 5) are summarized in table 1. Note that the cross-section was assumed to be circular.

Table 1: *Dimension for different Tesla valve designs*

name	$W_y^{[mm]}$	n	$Y3^{[mm]}$	$X2^{[mm]}$	$R^{[mm]}$	$BETA^{[\circ]}$	$ALPHA^{[\circ]}$	$LENOUT^{[mm]}$
Design A	1	0.990	0.600	1.5	2.50	8.53	45.0	2.00
Design B	1	0.797	0.608	1.6	2.35	71.1	41.9	2.94

2.1.2 Mesh

NOT COMPLETE

- show mesh for different designs A & B
- show naming convention

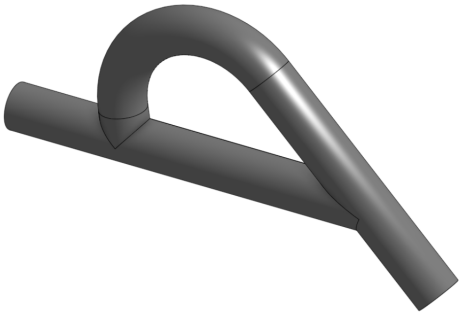


Figure 4: *Tesla valve design A used for this simulation.*



Figure 5: *Tesla valve design B used for this simulation.*

2.2 Numerical Model

2.2.1 Flow Equations

The flow equations are the Navier Stokes equations for incompressible flow, which in the strong form can be stated as:

$$\dot{u} + \nabla u \cdot u - \nu \Delta u + \nabla p = f \tag{2}$$

$$\nabla \cdot u = 0 \tag{3}$$

where u is the unknown velocity, p is the pressure field, ν is the given kinematic viscosity and f a given source. The Navier Stokes equations can be rewritten in the finite element weak form as:

$$F = \left(\frac{u - u_0}{dt}, v \right) + (\nabla u_m \cdot u_m, v) + \nu \left(\nabla u_m, \nabla v \right) + (\nabla p, v) + (\nabla \cdot u_m, q) - (f, v) \tag{4}$$

where $u_m = \theta u + (1 - \theta)u_0$ with $\theta = 0.5$ in the Crank-Nicholson scheme.

2.2.2 Stabilization methods

It is well known that the numerical solution of Navier-Stokes equations using Finite Element Methods suffers from two sources of numerical instabilities [4] [5]:

- Using wrong combination of velocity and pressure finite element spaces (that do not satisfy the inf-sup condition) generate numerical oscillations in the pressure field (spurious pressure).
- When the problem is convection-dominated ($Re >> 1$), numerical oscllations in the velocity field occurs (called spurious velocity).

These sources of instabilities may be controlled in two ways:

- Using appropriate combination of velocity and pressure finite element spaces.
- Adding stabilization terms to the weak formulation of the problem.

The first item can be addressed by using a 'P_k - P_{k+1}' combination of spaces for the unknown velocity and pressure fields. Such mixed formulation is expressed in FEniCS with the following lines of code, in which the Vector Element space $Q_e = P_1$ for the pressure and the space $V_e = P_2$ for the velocity and mixed in the space $W = V_e \times Q_e = P_k \times P_{k+1} = P_1 \times P_2$.

```
Ve = VectorElement("CG", mesh.ufl_cell(), 2)
Qe = FiniteElement("CG", mesh.ufl_cell(), 1)
W = FunctionSpace(mesh, Ve * Qe)

# Define test functions
w = Function(W)
u, p = split(w)
(v, q) = TestFunctions(W)
w0 = Function(W)
u0, _ = split(w0)
```

The second way to reduce numerical instabilities is to add to the weak formulation a stabilization term as expressed in [4], [5] and [6]. In our formulation the stabilization term is the following

$$F_{stabilization} = \tau \left[(\nabla p + \nabla u_m \cdot u_m - f, \nabla q + \nabla v \cdot u_m) + (\nabla \cdot u_m, \nabla \cdot v) \right] \tag{5}$$

where τ is the stabilization parameter defined as

$$\tau = \left[\left(\frac{2}{dt} \right)^2 + \left(\frac{2\,\|u\|}{h} \right)^2 \left(\frac{4\nu}{h^2} \right)^2 \right]^{(-0.5)} \tag{6}$$

in which h is defined as the mean mesh cell diameter.

2.2.3 Boundary Condition

The inlet velocity is assumed to have velocity only along the x-axis with a paraboloid profile with the following shape:

$$v_{in} = U(1 - \frac{y^2}{R^2} - \frac{z^2}{R^2}) \tag{7}$$

where R is the radius of the conduct and U is the maximum value of the velocity profile. Velocity components along y and z are supposed to have zero values. Dirichlet no-slip condition is applied on the walls of the valve and the outlet is assumed to discharge to atmospheric pressure.

2.2.4 Benchmark Problem

NOT COMPLETE

- explain 2d flow in around a cylinder
- Material/parameters
- what are we benchmarking

2.2.5 Numerical Analysis

NOT COMPLETE

- Simulation strategy:
  - 4 designs: designA, designA-reverse, designB, designB-reverse
  - use different velocities: low 0.3, medium 1 and high 3
- Material/parameters

3 Software Workflow

The CAD geometry is designed using OnShape[2]. An exported 3D *.step* file is imported into GmSH[3] to define the mesh and surface boundaries. The mesh is then exported as a *.msh* file and is converted to *.xdmf* domain and boundary files using meshio library[4]. The calculations are performed using FEniCS open source computing platform. ParaView[5] is used for the visualization and post-processing of the results.

[2] `https://www.onshape.com/`
[3] `https://gmsh.info/`
[4] `https://github.com/nschloe/meshio`
[5] `https://www.paraview.org/`

4 Results and discussion

NOT COMPLETE

- flow results
- pressure results
- diodicity discussion and comparison

References

[1] Kwang, Chong,emphA review of microvalves, J. Micromech. Microeng. 16 (2006)

[2] Jin-Yuan Qian, Cong-Wei Hou, Xiao-Juan Li and Zhi-Jiang Jin, *Actuation Mechanism of Microvalves: A Review*, Micromachines 2020, 11, 172

[3] A. Y. Nobakht, M. Shahsavan, A. Paykani, *Numerical Study of Diodicity Mechanism in Different Tesla-Type Microvalves*, Journal of Applied Research and Technology, Vol.11, December 2013

[4] T.E. Tezduyar, *Stabilized finite element formulations for incompressible flow computations*, Advances in Applied Mechanics, 28 (1992) 1-44

[5] T.E. Tezduyar, S. Mittal, S.E. Ray and R. Shih, *Incompressible flow computations with stabilized bilinear and linear equal-order-interpolation velocity-pressure elements*, Computer Methods in Applied Mechanics and Engineering, 95 (1992) 221-242

[6] J. Jansson, E. Krishnasamy, M. Leoni, N. Jansson and J. Hoffman, *Time-resolved Adaptive Direct FEM Simulation of High-lift Aircraft Configurations*

[7] A. Logg, H. P. Langtangen, *Solving PDEs in Python*, Springer 2017

[8] A. Logg, K. A. Mardal, G. N. Wells, *Automated Solution of Differential Equations by the Finite Element Method*, Springer 2017

[9] A. Gamboa, C. Morris, F. Foster, *Improvements in Fixed-Valve Micropump Performance Through Shape Optimization of Valves*, Journal of Fluids Engineering, March 2005, Vol. 127